

Electrolytic behavior and application to lithium batteries of monofluorinated dimethyl carbonate

Yukio Sasaki ^{a,*}, Masahiro Takehara ^a, Susumu Watanabe ^a, Masato Oshima ^a,
Noritoshi Nanbu ^a, Makoto Ue ^b

^a Department of Nanochemistry, Faculty of Engineering, Tokyo Polytechnic University, 1583 Iiyama, Atsugi, Kanagawa 243-0297, Japan

^b Mitsubishi Chemical Corporation, Science & Technology Research Center, 8-3-1 Chuo, Ami, Inashiki, Ibaraki 300-0332, Japan

Received 16 October 2005; accepted 21 October 2005

Abstract

Dimethyl carbonate (DMC) was fluorinated by F₂ gas (direct fluorination) to form four fluorinated DMC derivatives, such as monofluorinated DMC (MFDMC), two difluorinated DMCs (DFDMC(a) and DFDMC(b)) and trifluorinated DMC (TFDMC). The electrolytic behavior and charge–discharge characteristics of the lithium electrode were examined in MFDMC and MFDMC-ethylene carbonate (EC) equimolar binary solutions. The dielectric constant and viscosity of MFDMC are higher than those of DMC. This means that the molecular interaction in MFDMC becomes larger due to the introduction of a high electron withdrawing fluorine atom. In addition, oxidation durability of MFDMC is higher than that of DMC. The specific conductivity in EC-MFDMC equimolar binary solution containing 1 M LiPF₆ is 8.5 mS cm^{−1} at 25 °C which means a good electrolyte for lithium batteries. The lithium electrode cycling efficiency in EC-MFDMC equimolar binary solution containing 1 M LiPF₆ is higher than 70%. The surface of the film formed on the electrode in the solution is homogeneous, and consists of the uniform and small grain size. © 2005 Elsevier B.V. All rights reserved.

PACS: Batteries, 84.60.D; Electrochemistry 82.45

Keywords: Direct fluorination of dimethyl carbonate; Lithium batteries; Electrolytic conductivity; Cycling efficiency

1. Introduction

With increasing demand for power sources of mobile equipments and electric vehicles, it is becoming increasingly important to find a new solvent for lithium rechargeable batteries with high energy density, oxidation durability, high liquidus range, and non-flammability. One of the appropriate methods for finding a good performance solvent is the introduction of fluorine atoms into the solvent molecules. Among the important solvents for lithium batteries, fluorination using F₂ gas (direct fluorination) has already been applied to 1,2-dimethoxyethane (DME) [1], propylene carbonate (PC) [2] and ethylene carbonate (EC) [2]. In addition, we reported the formation of three monofluorinated γ -butyrolactones by direct fluorination, such as α -fluoro- γ -BL, $\hat{\alpha}$ -fluoro- γ -BL and γ -fluoro- γ -BL [3]. In general, direct fluorination is suitable to synthesize various fluorinated organic solvents.

The purpose of the present paper is to elucidate the electrolytic behavior and charge–discharge characteristics of lithium electrode in monofluorinated dimethyl carbonate (MFDMC) obtained by direct fluorination of dimethyl carbonate (DMC), which is well-known as a typical low viscous cosolvent with PC and EC for lithium batteries.

2. Experimental

F₂ gas was generated by the electrolysis of KF·2HF (Fluorodec™30, ToyoTanso Co. Ltd.) at 80–90 °C. Fluorination of DMC was carried out using 15% F₂ gas diluted by N₂ gas. The gas was poured onto the DMC sample surface at 22.4 ml min^{−1} with stirring in PFA vessel at 5 °C. The fluorinated DMC derivatives were confirmed by Gas Chromatograph and Mass Spectrometer (GC-MS)(JMS-SX102A). The purification of monofluorinated DMC (MFDMC) was carried out by fractional distillation of the fluorinated DMC sample. The purity of MFDMC was examined by elemental analysis (found: C, 33.4; H, 3.9; F, 17.5 wt.%; calculated: C,

* Corresponding author. Tel.: +81 46 242 9530; fax: +81 46 242 3000.

E-mail address: sasaki@nano.t-kougei.ac.jp (Y. Sasaki).

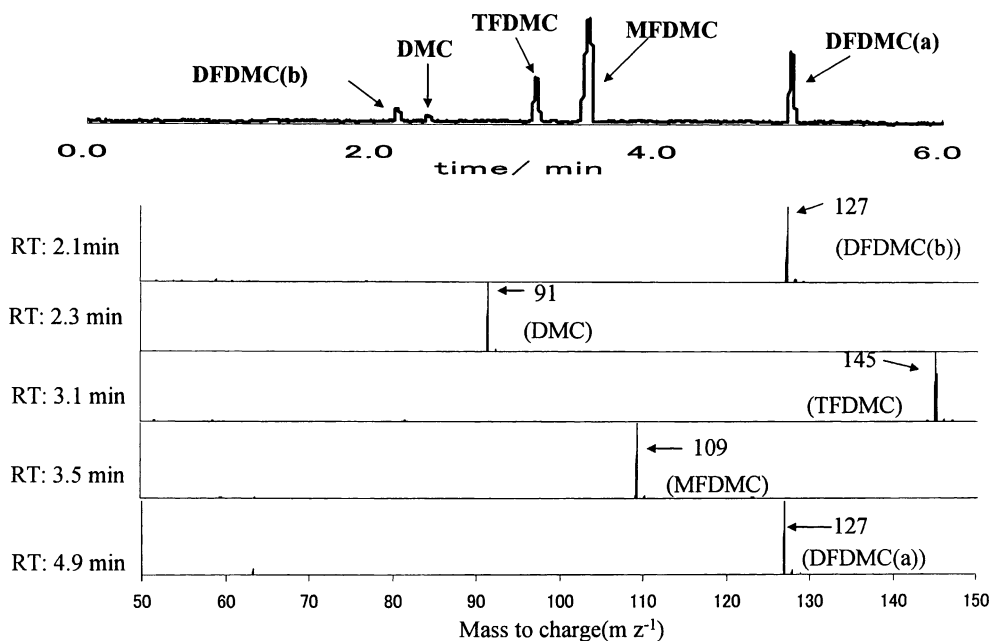


Fig. 1. GC-MS analysis of the fluorinated DMC derivatives after fluorination for 24 h.

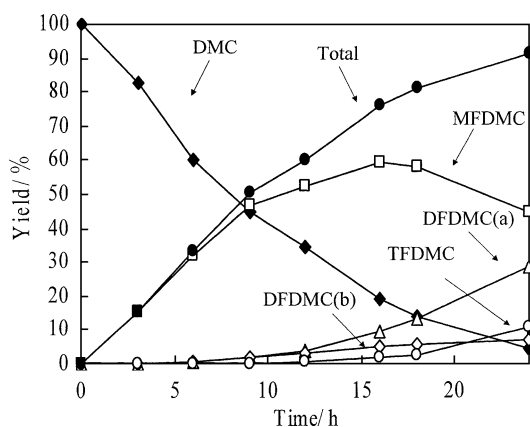


Fig. 2. Variation in yields with time of the fluorinated DMC derivatives.

33.3; H, 4.7; F, 17.6 wt.%) and ^1H and ^{13}C NMR (JEOL JNM-LA500). DMC and electrolytes were used as received (battery grade). MFDMC was dehydrated using purified molecular

sieves (4A) before preparation of the solutions. The preparation of the solution and the assembly of the cell were carried out in an argon-filled dry box. The apparatus and techniques for measurements of specific conductivity [4–6], dielectric constant [7] and viscosity [7] were reported previously. The lithium electrode cycling efficiency (charge–discharge coulombic efficiency of cycling for lithium electrode) was measured with a charge/discharge unit (Hokuto Denko, HJ-201B) and estimated by a galvanostatic plating–stripping method reported by Koch and Brummer [8]. The morphology of the films formed on a Ni (working) electrode after cycling was examined by using a scanning electron microscope (SEM, JSM-5310).

3. Results and discussion

Fig. 1 shows GC-MS analysis of the fluorinated DMC derivatives after fluorination for 24 h. In this figure, RT means the retention time. It is found that four fluorinated

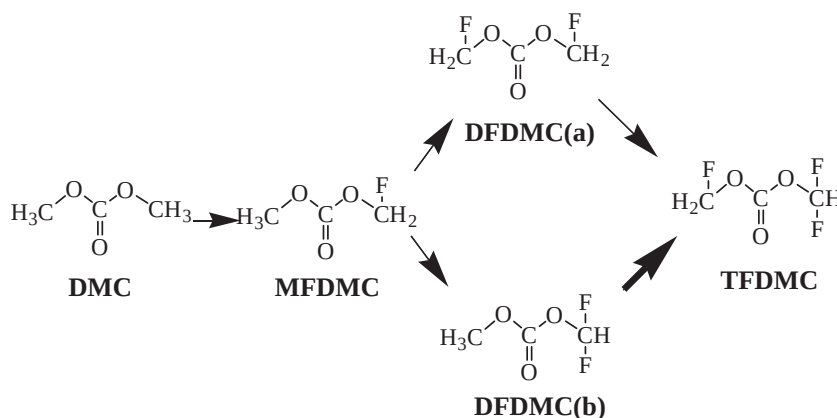


Fig. 3. Formation process of the fluorinated DMC derivatives.

Table 1

Correlation of HOMO and LUMO energies for fluorinated DMC derivatives calculated by B3LYP

	HOMO energy/kJ mol ⁻¹	LUMO energy/kJ mol ⁻¹
DMC	-790.8	6.9
MFDMC	-835.8	1.0
DFDMC(a)	-877.1	-37.2
DFDMC(b)	-873.3	-16.7
TFDMC	-881.1	-55.6

DMC derivatives are formed by direct fluorination of DMC, which are monofluorinated DMC (MFDMC), two difluorinated DMCs (DFDMC(a), DFDMC(b)), and trifluorinated DMC (TFDMC). Variation in yields with time of the fluorinated DMC derivatives is shown in Fig. 2. The initially formed product in Fig. 2 is MFDMC. The yield of MFDMC increases with decreasing DMC concentration. In addition, the yield decreases gradually after the maximum value at about 15 h. On the contrary, formation of two DFDMCs begins slowly after about 10 h, and then TFDMC formation begins successively. DFDMC(a) formation continues much to increase compared with that of DFDMC(b) after TFDMC formation. Accordingly, it seems that TFDMC formation mainly occurs through the DFDMC(b). From the total yield of the fluorinated DMC derivatives, it is found that fluorination of DMC occurs quantitatively. By considering the variations in the yields with time, formation process of the fluorinated DMC derivatives is assumed as shown in Fig. 3. Table 1 shows the HOMO and LUMO energies for fluorinated DMC derivatives calculated by B3LYP. In HOMO energies, oxidation durability decreases in the order, TFDMC>DFDMC(a)>DFDMC(b)>MFDMC>DMC. On the other hand, reduction durability increases in the same order as shown in LUMO energies. Therefore, it is found that the potential windows of the fluorinated DMC derivatives are shifted to oxidation sides. In the present study, pure MFDMC is obtained by distillation of the fluorinated sample after fluorination for 10 h because DFDMC(a) and DFDMC(b) formations are minor in that time. Fig. 4 shows *i*-*E* curves in MFDMC and DMC solutions. In Fig. 4, oxidation potential of MFDMC is slightly higher than that of DMC. On the other

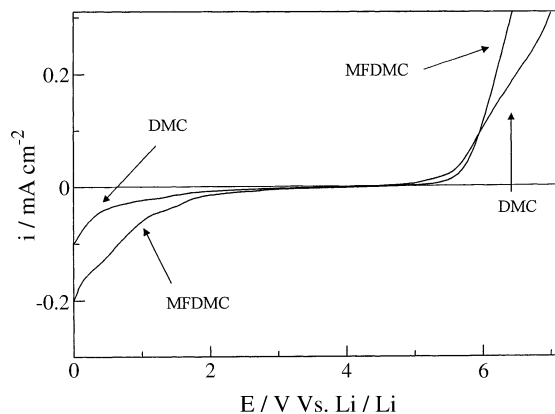


Fig. 4. *i*-*E* curves in 0.5 mol dm⁻³ LiBF₄ solution using a glassy carbon electrode at a scan rate of 5.0 mV s⁻¹ at 25 °C.

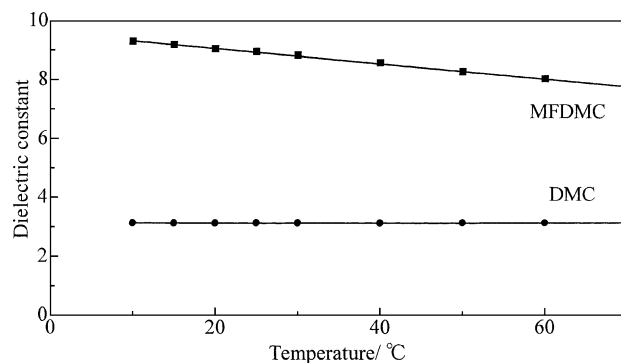


Fig. 5. Temperature dependence on dielectric constants of DMC and MFDMC.

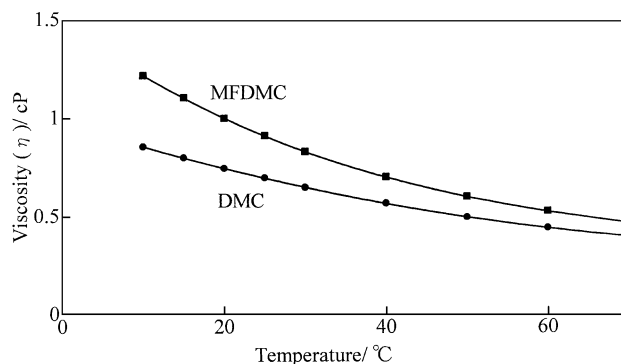


Fig. 6. Temperature dependence on viscosities (*η*) of DMC and MFDMC.

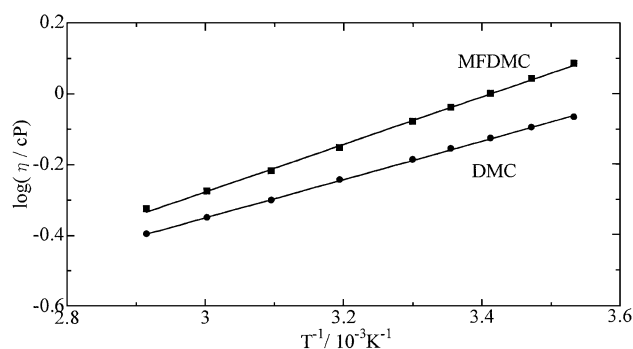


Fig. 7. Plot of log(*η*/cP) vs. 1/*T* for DMC and MFDMC.

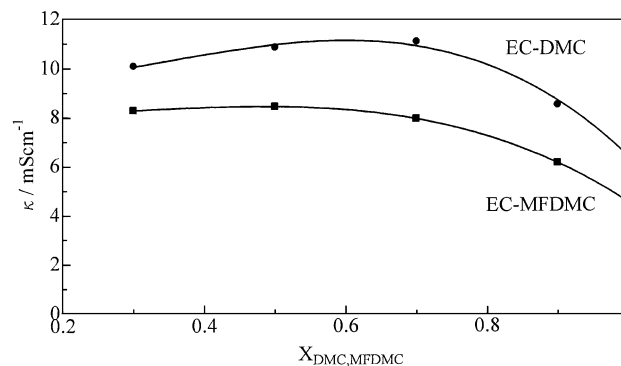


Fig. 8. Variation of specific conductivities in EC-DMC and EC-MFDMC binary solvent mixtures containing 1.0 M LiPF₆ at 25 °C. *X*_{DMC, MFDMC} means the mole fraction of DMC and MFDMC.

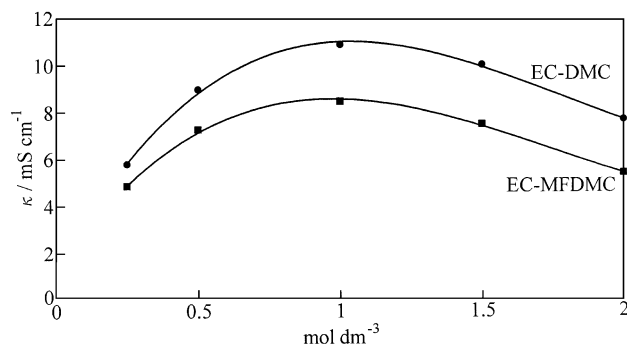


Fig. 9. Variation of specific conductivities (κ) with LiPF_6 concentrations in EC-DMC and EC-MFDMC equimolar binary solutions at 25 °C.

hand, reductive decomposition potential of MFDMC occurs easily compared with that of DMC. This fact is consistent with the HOMO and LUMO energies in Table 1.

Fig. 5 shows dielectric constants of MFDMC and DMC with temperature. As shown in Fig. 5, the dielectric constant of MFDMC is very high compared to DMC. The higher dielectric constant of MFDMC is dependent on the higher polarity of MFDMC due to the introduction of a high electron withdrawing fluorine atom. In addition, the dielectric constant of MFDMC decreases linearly with increasing temperature. This fact shows that the orientation of the dipoles in MFDMC reduces with a rise in temperature. On the other hand, the dielectric constant of DMC remains almost constant at whole temperature range.

Fig. 6 shows the variation of viscosities (η) for MFDMC and DMC with temperature. As expected by the dielectric constants, the viscosity of MFDMC is higher than that of

DMC. However, the viscosity of MFDMC is nearly equal to that of water at 25 °C. Plot of $\log \eta$ vs. $1/T$ for MFDMC and DMC is shown in Fig. 7. These plots give the straight lines. Activation energies of viscous flow estimated from these lines are 12.9 and 10.3 kJ mol⁻¹ for MFDMC and DMC, respectively. The activation energy of MFDMC with high viscosity becomes higher than that of DMC.

Fig. 8 shows the specific conductivities in EC-DMC and EC-MFDMC binary solvent mixtures containing 1.0 M LiPF_6 at 25 °C. EC-MFDMC solution shows the highest conductivity of 8.5 mS cm⁻¹ at mole fraction of 0.5. The conductivity is available for lithium batteries. The dependence on the LiPF_6 concentrations of the conductivities in EC-DMC and EC-MFDMC equimolar binary solutions is shown in Fig. 9. The conductivities of both solutions show the maximum values at 1 M LiPF_6 . In addition, the difference of conductivities between these solutions in Figs. 8 and 9 depends on the viscosities in Fig. 6. Fig. 10 shows the lithium electrode cycling efficiency in MFDMC, EC-DMC and EC-MFDMC equimolar binary solutions containing 1 M LiPF_6 at 25 °C (a), and SEM photographs on Ni electrode (working) in these solutions after discharged of the 10th cycle (b). In Fig. 10(a), EC-MFDMC solution shows the highest efficiency more than 70% in a higher range of cycle numbers. MFDMC solution shows a moderate efficiency in a range of 60th cycle. However, the efficiency in EC-DMC solution decreases rapidly with the cycle number. As a result, the EC-MFDMC solution shows a good electrolyte for rechargeable lithium batteries. On the other hand, lithium dendrites are formed on the electrode in EC-DMC solution in Fig. 10(b). Furthermore, the surface film on

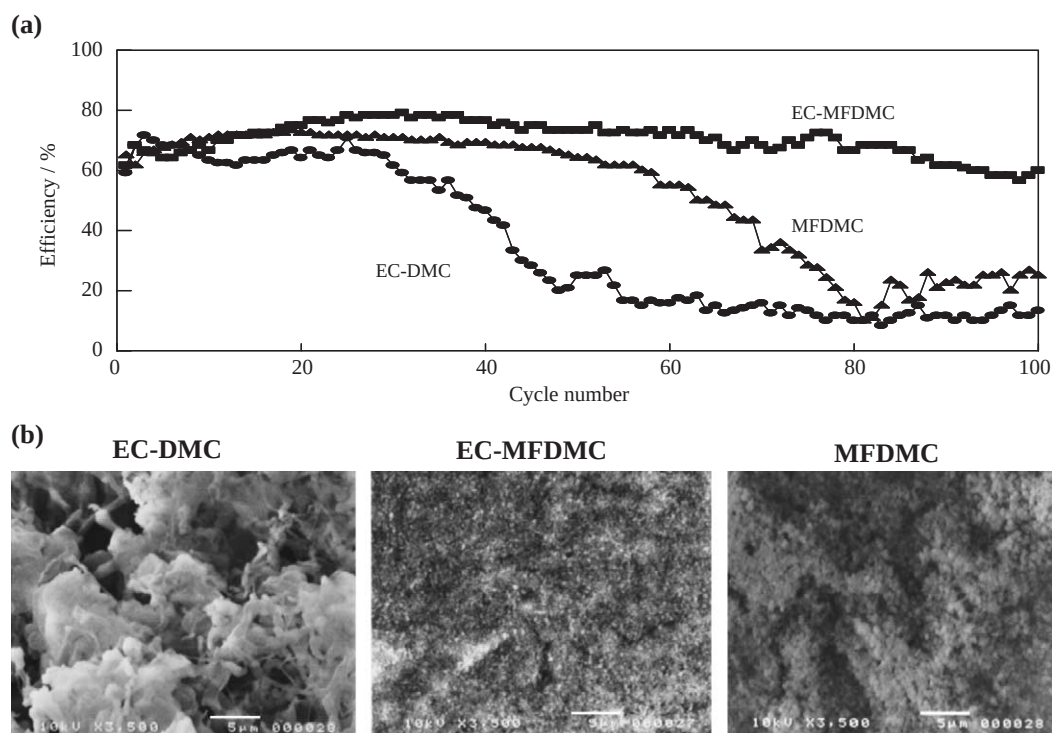


Fig. 10. Variation of lithium electrode cycling efficiencies ($i_p=i_s=1.0 \text{ mA cm}^{-2}$, $Q_p=0.3 \text{ C cm}^{-2}$) in MFDMC, EC-DMC and EC-MFDMC equimolar binary solutions containing 1.0 M LiPF_6 at 25 °C (a) and SEM photographs on Ni electrode in these solutions after discharged of the 10th cycle (b).

the electrode in MFDMC solution is somewhat heterogeneous. However, the surface film in EC-MFDMC solution is very homogeneous, and consists of the uniform and small grain size. Accordingly, the high cycling efficiency in EC-MFDMC solution is related to the morphology of the films, which are probably composed of the fluorinated compounds.

4. Conclusions

Four fluorinated DMC derivatives, such as MFDMC, DFDMC(a), DFDMC(b) and TFDMC are formed by direct fluorination of DMC. The dielectric constant and viscosity of MFDMC are higher than those of DMC. EC-MFDMC equimolar binary solution containing 1 M LiPF₆ shows higher specific conductivity of 8.5 mS cm⁻¹ at 25 °C, which means a good electrolyte for lithium batteries. The lithium electrode cycling efficiency (charge–discharge coulombic efficiency of cycling for lithium electrode) in EC-MFDMC equimolar binary

solution containing 1 M LiPF₆ is higher than 70%. The surface of the film formed on the electrode in the solution is homogeneous, and consists of the uniform and small grain size.

References

- [1] J.L. Adcock, R.J. Lagow, *J. Org. Chem.* 38 (1973) 3617.
- [2] J. H. Prager, US Patent 3, 455, 954 (1969).
- [3] Y. Sasaki, R. Ebara, N. Nanbu, M. Takehara, M. Ue, *J. Fluorine Chem.* 108 (2001) 117.
- [4] M. Handa, S. Fukuda, Y. Sasaki, K. Usami, *J. Electrochem. Soc.* 144 (1997) L235.
- [5] M. Handa, M. Suzuki, J. Suzuki, H. Kanematsu, Y. Sasaki, *Electrochem. Solid-state Lett.* 2 (1999) 60.
- [6] Y. Sasaki, S. Sekiya, M. Handa, K. Usami, *J. Power Sources* 79 (1999) 91.
- [7] Y. Sasaki, K. Miyagawa, N. Wataru, H. Kaido, *Bull. Chem. Soc. Jpn.* 66 (1993) 1608.
- [8] V.R. Koch, S.B. Brummer, *Electrochim. Acta* 23 (1978) 55.